GGCT-Net: A Dual-Branch Gated Graph Convolution and Grouped Cross-Attention Transformer Network for Hyperspectral Unmixing

Qingfei Liu, Xiaodong Yu, Hongbin Dong, and Shuying Zang

***Abstract*—In recent years, deep learning-based hyperspectral unmixing (HU) techniques have garnered increasing attention and achieved significant progress. However, existing deep learning methods exhibit notable limitations: convolutional neural networks (CNNs) struggle to correlate spectrally similar yet spatially dispersed pixels, while Transformers lack the ability to distinguish between spectral and spatial features when processing high-dimensional data. To address these, this article proposes a dual-branch gated graph convolution and grouped cross-attention Transformer network (GGCT-Net). Specifically, this network designs a spatial–spectral cross-attention block (SCAB) within the grouped cross-attention Transformer (GCA-Transformer) encoder to process spectral and spatial features separately. Within the gated graph convolutional network (G-GCN) encoder, a graph structure based on spectral similarity is constructed to correlate spectrally similar yet spatially dispersed pixels, and deep feature interactions are achieved through a gated cross-attention fusion (GCAF) module. The experimental results on both simulated and real hyperspectral datasets demonstrate the proposed method's effectiveness and superiority compared to various HU approaches.**

***Index Terms*—Gate-control, graph convolution network (GCN), hyperspectral unmixing (HU), spatial–spectral features, Transformer.**

I. Introduction

HYPERSPECTRAL imaging technology [1] acquires continuous narrowband information of terrestrial objects with nanometer-level spectral resolution, forming a "map-spectrum integration" data cube [2] that enables precise identification based on the unique spectral "fingerprints" of substances. However, due to limitations in sensor resolution [3] and the complex distribution of natural objects, "mixed pixels" [1] are commonly present in images, thereby restricting classification and quantitative inversion accuracy. Consequently, hyperspectral unmixing (HU) [4] has become a critical processing step. The objective of HU is to decompose mixed pixels into "endmembers" representing pure substances and their corresponding "abundances." This process serves as the core technology for subpixel-level information extraction. This technique has been shown to significantly enhance the application value of hyperspectral data, and has been adopted in a variety of critical fields, including geological exploration [5], precision agriculture [6], and environmental monitoring [7]. The technique provides in-depth information support for related decision-making.

Among HU methods, the linear mixture model (LMM) [8] is a widely adopted technique due to its simplicity, efficiency, and ability to effectively describe spectral mixing processes. This model employs a representation of mixed pixels as linear combinations of endmember spectra and their respective abundances. Based on LMM, existing approaches primarily fall into three categories: geometric methods, statistical methods, and sparse regression methods. In the field of geometric methods, vertex component analysis (VCA) [9] and fully constrained least squares unmixing (FCLSU) [10] are the most frequently employed; statistical methods are typified by nonnegative matrix factorization (NMF) [11]. In order to surmount the nonuniqueness of its solutions, researchers have introduced prior knowledge in the form of smoothness, sparsity, or geometric constraints, developing improved algorithms, such as minimum volume constrained NMF (MVC-NMF) [12]. Sparse regression methods are predicated on the principles of sparse representation theory. Representative algorithms, such as the SUnSAL series [13], address constrained sparse regression problems by automatically selecting relevant endmembers from extensive dictionaries, rendering them appropriate for unmixing tasks in scenarios where endmembers are unknown or the environment is complex.

Traditional linear mixing models (LMMs), such as the standard FCLSU, often exhibit limitations in addressing nonlinear effects [14] and spectral variability [15] in HU due to their reliance on manually designed prior knowledge. Although some extended linear methods attempt to incorporate variability modeling, they still face significant challenges. In recent

Received 10 November 2025; revised 20 January 2026; accepted 23 February 2026. Date of publication 26 February 2026; date of current version 6 March 2026. This work was supported in part by the State Key Program of National Natural Science of China under Grant 42430412 and in part by the Open Research Fund of Heilongjiang Key Laboratory of Analysis on Machine Learning and Dynamic System under Grant HLJKL2503. *(Corresponding author: Xiaodong Yu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3668181

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years, deep learning methods have emerged as a research frontier in this field, leveraging their powerful feature extraction and nonlinear modeling capabilities. Among these, convolutional neural network (CNN) [16] based approaches serve as pioneers and dominant solutions. Early studies established the foundation of deep unmixing by employing multilayer perceptrons or autoencoders. Subsequently, more sophisticated and specialized CNN architectures have been proposed: residual networks (ResNet) [17] alleviate gradient vanishing through skip connections, enabling the learning of deeper spectral features; U-Net and its variants [18] utilize an encoder–decoder structure to achieve pixel-level abundance inversion, effectively capturing spatial distribution patterns; 3D-CNNs [19] are directly applied to hyperspectral cubes, simultaneously extracting joint spectral–spatial features from adjacent pixels, which significantly improves unmixing accuracy. Furthermore, deep convolutional autoencoders (DCAEs) [20] learn abundances through the encoder and treat the decoder weights as an endmember dictionary to reconstruct spectral signals; multiscale convolutional networks [21] capture spatial contexts at varying ranges using parallel or sequential convolutions with different kernel sizes, and the integration of convolutional networks with fully convolutional networks (FCNs) [22] enables end-to-end abundance map generation. Collectively, these methods demonstrate the strong potential of CNNs in exploiting local correlations within hyperspectral data.

Subsequent to CNNs, Transformer-based unmixing methods have exhibited considerable potential. The Transformer's core component, the self-attention mechanism, facilitates global computation of dependencies between pixels or spectral bands, thereby circumventing the constraints imposed by CNNs' local receptive fields. Researchers have creatively applied this powerful architecture for HU, developing multiple innovative models. The fundamental concept of the vision Transformer (ViT) [23] is applied directly in this context: hyperspectral pixels or image patches are treated as sequences, with self-attention layers learning global spectral features to achieve excellent endpoint representation capabilities. To overcome the reliance of ViT on large-scale data pretraining and better integrate spatial information, the spectral Transformer [24] was proposed. The model's primary function is to identify discriminative spectral sequence features through group embeddings and cross-layer attention connections. Addressing the high computational complexity of Transformers, the patch Transformer [25] divides images into overlapping or nonoverlapping patches, performing computations at the patch level to effectively balance performance and efficiency. Furthermore, the hierarchical Transformer [26] constructs a pyramid-like multiscale architecture to capture features from global-to-local scales at different resolutions. The masked autoencoder (MAE) [27] paradigm is also applied to the unmixing task, where the model is driven to learn robust spectral representations by reconstructing randomly masked spectral bands.

However, despite significant advances in deep learning methodologies based on CNNs and Transformers, these methods encounter several critical challenges and limitations in HU tasks. On the one hand, CNN-based unmixing methods primarily rely on the spatial adjacency of pixels, neglecting the intrinsic spectral similarity [28] connections between them. This approach, predicated on local spatial neighborhoods, encounters difficulties in effectively capturing correlations between pixels that are highly similar in spectral features but spatially distant. This limitation restricts the methods' ability to understand complex feature distributions and subtle spectral variations. On the other hand, the majority of transformer-based unmixing methods utilize a unified multihead attention mechanism to concurrently process spatial and spectral characteristics. However, hyperspectral data possesses a large number of channels with properties that differ significantly from the spatial dimension. This "one-size-fits-all" approach to attention is ill-equipped to effectively model heterogeneous features and is highly susceptible to an exponential increase in computational complexity.

In order to address the aforementioned challenges, this article proposes a novel HU neural network, gated graph convolution, and grouped cross-attention Transformer network (GGCT-Net), which achieves efficient feature extraction through the parallel integration of a gated graph convolutional network (G-GCN) and a grouped cross-attention Transformer (GCA-Transformer). The primary benefits of this model are as follows. First, the GCA-Transformer module employs a dual-path attention mechanism to independently extract features from both the channel and spatial dimensions. The channel path is oriented toward the preservation of endmember authenticity by means of spectral correlations, whilst the spatial path is directed toward the enhancement of abundance spatial continuity by capturing long-range dependencies. The integration of features from both approaches ultimately results in a more comprehensive representation. Second, the G-GCN module constructs a graph structure based on spectral feature similarity, independent of spatial location. This effectively establishes semantic associations by connecting spectrally matched but spatially separated pixels. Concurrently, it dynamically adjusts the similarity metric through an adaptive Gaussian kernel bandwidth, enabling the graph structure to better align with actual object distributions. This enhancement to the feature's discriminative power is accompanied by an adaptation to irregular superpixel regions.

The main contributions of this article are summarized as follows.

1) We propose a novel HU network, GGCT-Net. The model employs a dual-branch encoder composed of a superpixel-guided G-GCN and a GCA-Transformer, which process local and global spatial–spectral features to achieve effective complementarity between local and global information, respectively.
2) We propose a G-GCN and design a gated cross-attention fusion (GCAF) module, which facilitates bidirectional deep interaction between the endmember and abundance branches via an adaptive gating mechanism. Specifically, the abundance estimation branch provides spatial constraints to enhance the authenticity of endmembers, while the endmember representation branch injects semantic features to improve the accuracy of abundance estimation. This fusion mechanism dynamically regulates the contribution of different feature sources,

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enabling refined feature selection and weighted fusion, thereby significantly improving the overall accuracy of both endmember extraction and abundance estimation.

3) We propose the GCA-Transformer and design the spatial–spectral cross-attention block (SCAB) to explicitly decouple feature processing across spectral and spatial dimensions. This design achieves fine-grained attention mechanisms and cross-dimensional feature fusion, ultimately constructing a more global and effective spectral–spatial integrated representation for HU.

The remainder of this article is organized as follows. Related work is described in Section II. The proposed method is introduced in Section III. Section IV details our experimental results on hyperspectral datasets. Finally, Section V concludes this article.

II. Related Works

A. Data Segmentation

In order to leverage spectral similarity in an effective manner, pixels that are both spatially contiguous and spectrally similar are aggregated into consistent units, i.e., superpixels. This article introduces the simple linear iterative clustering (SLIC) method [29]. This method is both simple and efficient, generating compact and uniform superpixels, making it suitable for presegmenting homogeneous regions in HU.

The standard SLIC was originally designed for color images. To address the characteristics of hyperspectral images, their core concept must be extended. The objective is to construct a comprehensive distance metric that considers both spatial proximity and spectral similarity. Assuming a hyperspectral image has L-bands, we represent each pixel i as a feature vector $[\mathbf{l}_i, \mathbf{c}_i]^T$, where $\mathbf{l}_i = [x_i, y_i]^T$ denotes its 2-D spatial coordinates, $\mathbf{c}_i \in \mathbb{R}^{L\times 1}$ denotes its spectral vector. Correspondingly, each cluster center (seed point) $\mathbf{C}_k$ is similarly represented as $[\mathbf{l}_k, \mathbf{c}_k]^T$. Then, the composite distance D' between pixel i and cluster center $\mathbf{C}_k$ is defined as

$$D' = \sqrt{\left(\frac{d_s}{S}\right)^2 + \left(\frac{d_c}{m}\right)^2} \tag{1}$$

where d_s represents the spatial distance, defined as $d_s = \|\mathbf{l}_i - \mathbf{l}_k\|_2$, d_c denotes the spectral distance, used to quantify the similarity between two spectral vectors $\mathbf{c}_i$ and $\mathbf{c}_k$. S denotes the maximum expected intraclass spatial distance, defined as $S = (N/K)^{1/2}$, where N is the total number of pixels, and K is the predefined number of superpixels. The compactness factor m balances the relative importance of spatial and spectral distances in the composite metric.

However, the performance of SLIC depends on the appropriate selection of hyperparameters, particularly the number of superpixels K and the compactness factor m. Suboptimal choices may lead to over-segmentation or under-segmentation, thereby misrepresenting homogeneous regions. Despite this, SLIC remains an effective preprocessing step that provides foundational support for constructing graph structures in the subsequent G-GCN. Its limitations also motivate further exploration of adaptive segmentation methods better suited to the characteristics of hyperspectral data.

B. Graph Convolution Network

The operational mechanism of GCNs [30] can be understood through the lens of spectral graph theory. Analogous to the Fourier transform, the convolution operation on the graph is defined as the product of the signal and the filter in the spectral domain. Based on the first-order approximation simplification using Chebyshev polynomials, the propagation rules for GCN layers are as follows:

$$\mathbf{H}^{(l+1)} = \sigma\left(\hat{\mathbf{D}}^{-\frac{1}{2}}\hat{\mathbf{A}}\hat{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l)}\mathbf{W}^{(l)}\right) \tag{2}$$

where $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}_N$ denotes the adjacency matrix augmented with self-connections, $\mathbf{I}_N$ is the identity matrix. $\hat{\mathbf{D}}$ is the degree matrix of $\hat{\mathbf{A}}$, a diagonal matrix where the diagonal elements $\hat{\mathbf{D}}_{ii} = \sum_j \mathbf{A}_{ij}$. $\mathbf{H}^{(l)} \in \mathbb{R}^{N\times F}$ represents the node feature matrix for layer l. $\mathbf{W}^{(l)} \in \mathbb{R}^{F\times F'}$ denotes the trainable weight parameter matrix for layer l. $\sigma(\cdot)$ denotes a nonlinear activation function, such as ReLU.

The application of GCN–HU tasks has been demonstrated to offer significant advantages. It is posited that each pixel can be regarded as a graph node, with the node feature being the spectral vector $\mathbf{x}_i \in \mathbb{R}^{L\times 1}$. The adjacency matrix $\mathbf{A}$ can be constructed by calculating spectral similarity between nodes and incorporating a spatial k-nearest neighbor strategy

$$\mathbf{A}_{ij} = \begin{cases} \text{sim}\left(\mathbf{x}_i, \mathbf{x}_j\right), & \text{if } \mathbf{x}_j \in \mathcal{N}_k\left(\mathbf{x}_i\right) \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where $\text{sim}(\cdot,\cdot)$ denotes the similarity metric function, and $\mathcal{N}_k(\mathbf{x}_i)$ represents the set of k nearest neighbors of node i in the space.

C. Gate Control Mechanism

Gate mechanisms [31] represent a crucial feature regulation approach in deep learning, dynamically controlling information flow within neural networks through learnable gate signals. In contradistinction to conventional static weights, this apparatus facilitates the adaptive selection of salient information and the suppression of redundant or noisy data based on input features. Consequently, this enhances the network's capacity to execute complex tasks.

The most fundamental mathematical representation of a gating mechanism is the gated linear element, which can be expressed in its basic form as follows:

$$\mathbf{h} = \mathbf{x} \otimes \sigma\left(\mathbf{W}_g\mathbf{x} + \mathbf{b}_g\right) \tag{4}$$

where $\mathbf{x}$ is the input feature vector, and $\mathbf{W}_g$ and $\mathbf{b}_g$ represent the gate weight matrix and bias term, respectively.

In scenarios involving more complex feature fusion, gating mechanisms are frequently employed to regulate interactions between features from disparate sources or levels. In the event of two feature representations, $\mathbf{F_1}$ and $\mathbf{F_2}$, requiring fusion, a gating-based fusion strategy can be expressed as follows:

$$\mathbf{g} = \text{Sigmoid}\left(\mathbf{W}_f\left[\mathbf{F_1}\|\mathbf{F_2}\right] + \mathbf{b}_f\right) \tag{5}$$

$$\mathbf{F}_{\text{fused}} = \mathbf{g} \otimes \mathbf{F_1} + (1 - \mathbf{g}) \otimes \mathbf{F_2} \tag{6}$$

where $[\cdot\|\cdot]$ denotes the feature concatenation operation, and $\mathbf{g}$ is the fusion gate vector. This mechanism enables the network

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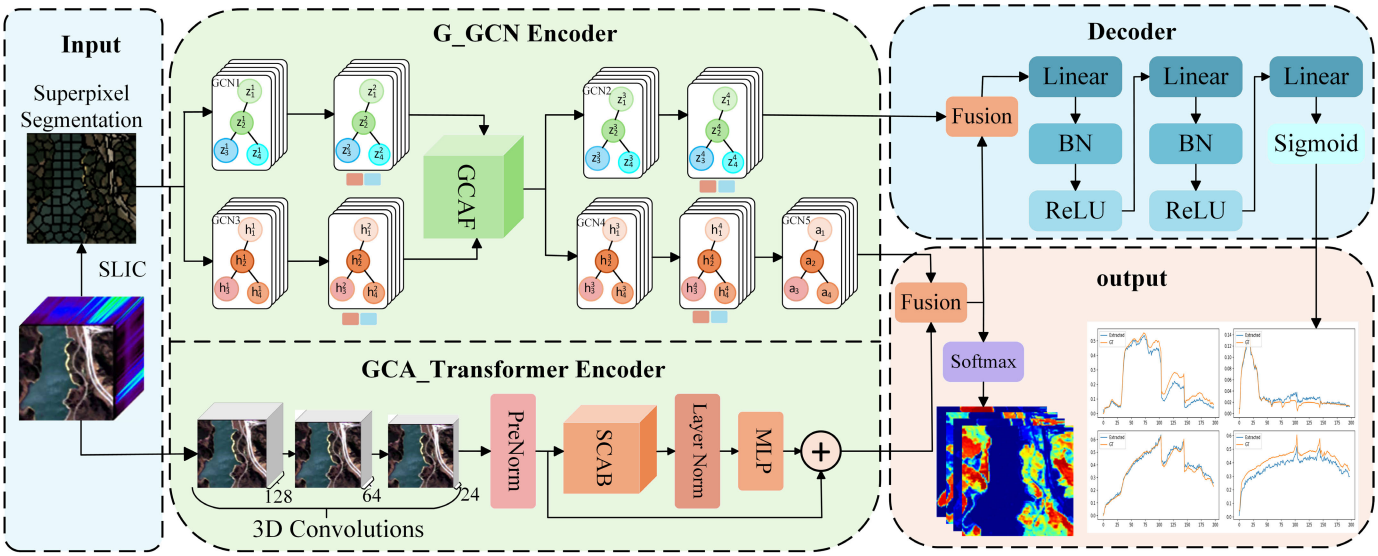


Fig. 1. Overall architecture of GGCT-Net. The decoder section illustrates the detailed structure of a single branch, while the complete decoder consists of P identical branches operating in parallel.

to autonomously determine whether to retain more information from $\mathbf{F_1}$ or $\mathbf{F_2}$ in each feature dimension, achieving soft feature selection and weighted fusion.

Another significant variant of the gating mechanism is the gated attention network, which integrates the gating concept with an attention mechanism. In the context of a query vector $\mathbf{q}$ and a key–value pair $(\mathbf{K}, \mathbf{V})$, gated attention can be expressed as follows:

$$\alpha_i = \frac{\exp\left(\mathbf{q}^T \mathbf{k}_i / \sqrt{d}\right)}{\sum_j \exp\left(\mathbf{q}^T \mathbf{k}_j / \sqrt{d}\right)} \tag{7}$$

$$g_i = \sigma\left(\mathbf{W}_a \left[\mathbf{q} \| \mathbf{k}_i\right] + \mathbf{b}_a\right) \tag{8}$$

$$\mathbf{h} = \sum_i \left(g_i \otimes \alpha_i\right) \mathbf{v}_i. \tag{9}$$

The present formula introduces an additional gating factor, designated g_i, in addition to the standard attention weight, denoted α_i. This factor modulates the contribution of each attention head or position to the final output, thereby enhancing the model's expressive power and robustness.

III. Proposed Method

In Section III, a comprehensive exposition of the GGCT-Net method is presented. The overall architecture of the network is illustrated in Fig. 1.

This network encoder adopts a dual-branch architecture: the G-GCN encoder and the GCA-Transformer encoder, which extract local spatial–spectral features and global contextual information, respectively. The G-GCN encoder is composed of two branches: endmember generation and abundance estimation. The former learns endmember distributions on superpixel maps, while the latter directly predicts abundance to maintain local consistency. The feature interaction is achieved through a gated cross-attention fusion module. The GCA-Transformer encoder first performs preliminary feature extraction on the raw image through convolution. It then captures long-range dependencies and global spectral–spatial features via a spatial–spectral cross-attention module combined with layer normalization and a feedforward network, compensating for the limitations of local information. Finally, the outputs from both encoders are fused to generate abundance, which is then fed into the decoder for further endmember generation.

A. Gated Graph Convolution Encoder

The gated graph convolution encoder uncovers local spectral–spatial correlations within hyperspectral data. It uses a dual-branch graph convolution network to learn terminal and abundance features, respectively, and introduces a gate-cross attention fusion module to facilitate information exchange between the two branches. Based on the GCN architecture, the encoder constructs a superpixel-level graph structure to model spatial correlations between pixels. First, the graph structure is initialized. Each region obtained from superpixel segmentation is treated as a node set V (with N nodes). Spectral similarity between nodes is computed using cosine distance to generate an adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$

$$\mathbf{A}_{ij} = \frac{\mathbf{X}_i \cdot \mathbf{X}_j}{\|\mathbf{X}_i\|_2 \left\|\mathbf{X}_j\right\|_2 + \varepsilon} \tag{10}$$

where $\mathbf{X}_i$ and $\mathbf{X}_j$ denote the spectral eigenvectors of nodes i and j, respectively, and ε represents the smoothing term to prevent division by zero. For the optimized Laplacian representation of the graph, the adjacency matrix undergoes normalization to yield the normalized graph Laplacian matrix

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}. \tag{11}$$

Here, $\mathbf{I}$ denotes the identity matrix, $\mathbf{D}$ denotes the degree matrix (satisfying $\mathbf{D}_{ii} = \sum_j \mathbf{A}_{ij}$).

Subsequently, it enters the graph convolutional layer for feature extraction. In order to enhance feature expression capabilities, residual connections and layer normalization are

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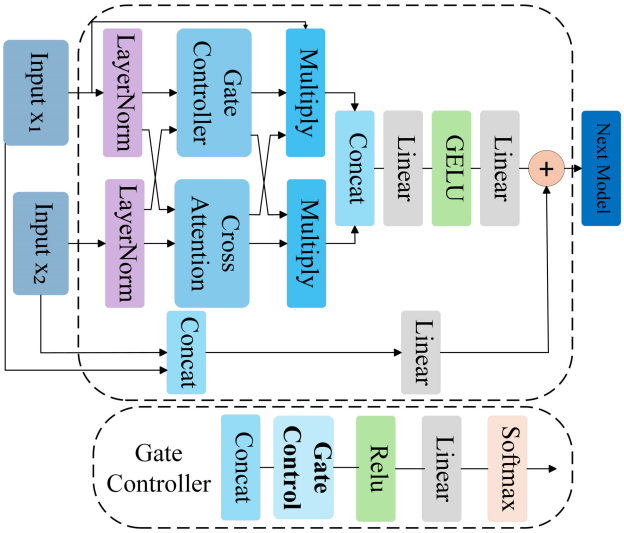


Fig. 2. Process of GCAF module.

introduced after each graph convolutional layer. This ensures stable gradient flow while improving model training stability.

The GCAF module represents a pivotal component in the integration of fine-grained features within this branch. The module under discussion enables deep feature interaction between the terminal branch and the abundance branch by means of cross-attention and adaptive gating mechanisms. As demonstrated in Fig. 2, the computational flow of the GCAF module is as follows.

First, layer normalization must be performed on each input feature

$$\tilde{\mathbf{X}}_1 = \text{LayerNorm}(\mathbf{X}_1) \tag{12}$$

$$\tilde{\mathbf{X}}_2 = \text{LayerNorm}(\mathbf{X}_2). \tag{13}$$

Subsequent to the process of normalization, the features are subjected to linear projection in order to generate the three components of attention. In this process, the query vector is derived from one feature stream, while the key and value vectors originate from another feature stream. Subsequently, the cross-attention weights are computed

$$\text{Attention}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{14}$$

$$\mathbf{A} = \text{MultiHeadAttention}\left(\tilde{\mathbf{X}}_1, \tilde{\mathbf{X}}_2, \tilde{\mathbf{X}}_2\right). \tag{15}$$

Softmax normalization occurs along the feature dimension, thereby ensuring that the sum of weights in each row is equal to one.

In order to achieve dynamic adjustment of the fusion ratio between dual features, GCAF incorporates a gating unit for the generation of fusion coefficients

$$\mathbf{G} = \text{Softmax}\left(\mathbf{W}_2 \cdot \text{ReLU}\left(\mathbf{W}_1\left[\tilde{\mathbf{X}}_1 \| \tilde{\mathbf{X}}_2\right] + \mathbf{b}_1\right) + \mathbf{b}_2\right) \tag{16}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are the learnable parameters, and $\|$ denotes the concatenation. The obtained gated weights $\mathbf{G} \in \mathbb{R}^{N\times 2}$ are used to balance the importance of the original features and the attention-enhanced features

$$\mathbf{X}_{\text{weighted}} = G_{i,0} \odot \tilde{\mathbf{X}}_1 \tag{17}$$

$$\mathbf{A}_{\text{weighted}} = G_{i,1} \odot \mathbf{A}. \tag{18}$$

In order to further enhance the expressive power of features, feature enhancers were also introduced

$$\mathbf{F} = \mathbf{W}_4 \cdot \text{GELU}\left(\mathbf{W}_3\left[\mathbf{X}_{\text{weighted}} \| \mathbf{A}_{\text{weighted}}\right] + \mathbf{b}_3\right) + \mathbf{b}_4 \tag{19}$$

where $\mathbf{W}_3$ and $\mathbf{W}_4$ are the learnable parameters. GELU stands for the Gaussian error linear unit activation function.

Finally, introducing residual connections ensures training stability

$$\mathbf{Y} = \mathbf{F} + \mathbf{W}_5\left[\mathbf{X}_1 \| \mathbf{X}_2\right]. \tag{20}$$

Through this sophisticated gating mechanism, the GCAF module adaptively adjusts the contribution of different feature sources. The fusion of complementary information across branches is achieved while preserving the integrity of the original information, thereby providing a more robust feature representation for subsequent unmixing tasks.

B. GCA-Transformer Encoder

The GCA-Transformer encoder has been developed for the purpose of capturing global long-range spectral–spatial dependencies across hyperspectral data. This addresses the limitations of the G-GCN branch in modeling long-range dependencies. The stage of the feature preprocessing is the extraction of shallow features from the input hyperspectral cube $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, via 3×3 convolutions and BatchNorm

$$\mathbf{X}_c = \text{BatchNorm}\left(\text{Conv}_{3\times 3}(\mathbf{X})\right). \tag{21}$$

Obtain $\mathbf{X}_c \in \mathbb{R}^{H\times W\times C}$.

The SCAB is pivotal to modeling global dependencies. The module under discussion captures long-range dependencies in both spectral and spatial dimensions through parallel spectral and spatial attention paths, respectively, then cross-fuses this multidimensional information, as demonstrated in Fig. 3. Below, we will provide a detailed explanation using the spectral attention path as an example.

Spectral attention path (modeling interband correlation). First, it is imperative to implement a channel grouping strategy that will facilitate the division of the C dimensional channels into G groups. Each group should comprise C/G channels

$$\mathbf{X}_{\text{channel}} = [\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_G], \quad \mathbf{X}_i \in \mathbb{R}^{B\times N\times (C/G)}. \tag{22}$$

For each grouped subfeature $\mathbf{X}_i$, the spatial dimension $H \times W \to 1\times 1$ is compressed via 1×1 convolution, yielding spectral dimension features

$$\mathbf{X}_s^g = \text{Conv}_{1\times 1}(\mathbf{X}_i). \tag{23}$$

A linear projection on $\mathbf{X}_s^g$ is required in order to generate a query, key, and value matrix for measuring spectral dependence between pixels

$$\mathbf{Q}_s^g = \mathbf{X}_s^g \mathbf{W}_s^q \tag{24}$$

$$\mathbf{K}_s^g = \mathbf{X}_s^g \mathbf{W}_s^k \tag{25}$$

$$\mathbf{V}_s^g = \mathbf{X}_s^g \mathbf{W}_s^v \tag{26}$$

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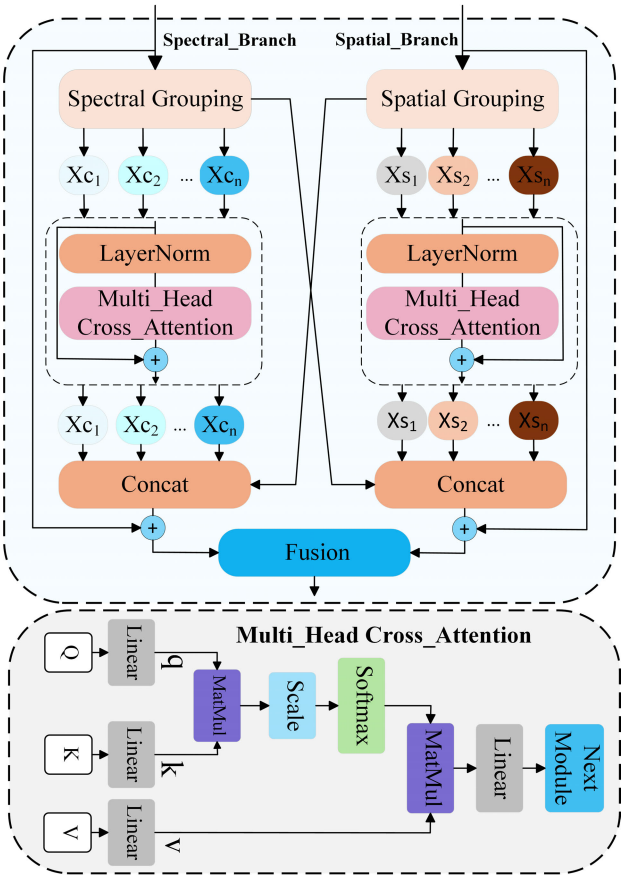


Fig. 3. Process of SCAB module.

where $\mathbf{W}_s^q, \mathbf{W}_s^k, \mathbf{W}_s^v \in \mathbb{R}^{(C/G)\times d}$ (d is the dimension of the attention head) are learnable weight matrices. Then, feature weighting is performed

$$\mathbf{ChnAttn} = \mathrm{Attn}_s^g \mathbf{V}_s^g. \tag{27}$$

Finally, the attention outputs from all channel groups are concatenated to form the complete channel attention feature

$$\mathbf{X}_{\mathrm{chn}}^{\mathrm{out}} = [\mathbf{ChnAttn}_1 \| \mathbf{ChnAttn}_2 \| \cdots \| \mathbf{ChnAttn}_G]. \tag{28}$$

The final output of dual-path attention is achieved through weighted summation

$$\mathbf{X}_{\mathrm{trans}}^{\mathrm{out}} = \alpha \cdot \mathbf{X}_{\mathrm{chn}}^{\mathrm{out}} + \beta \cdot \mathbf{X}_{\mathrm{spa}}^{\mathrm{out}} \tag{29}$$

where α and β represent learnable weight parameters. These parameters are subject to automatic optimization through a process known as backpropagation.

Ultimately, the output from the GCA-Transformer branch is restored to its original spatial resolution through an upsampling operation, and its features are deeply fused with those from the G-GCN branch at the superpixel level. This grouped multidimensional cross-attention mechanism has been demonstrated to be effective in capturing global contextual information, whilst concomitantly enhancing the modeling capability of hyperspectral data's inherent characteristics through a dual-path design. This provides robust support for precise endmember extraction and abundance estimation.

C. Decoder

As the core module for "feature reconstruction and physical constraint enforcement," the decoder receives the fused features from both the G-GCN and GCA-Transformer encoders. To enable the parallel generation of multiple endmember spectra, the decoder adopts a parallel architecture with P independent branches, each corresponding to one endmember. Each branch internally employs a stacked progressive structure of "Linear-BN-ReLU" layers, with a final sigmoid activation function. Fig. 1 clearly annotates three linear layers, two batch normalization (BN) layers, two ReLU layers, and the final sigmoid activation function. The outputs of all branches are stacked along the endmember dimension to form the endmember spectral tensor.

D. Loss Function

This article uses a multitask loss function, which comprises reconstruction loss and KL divergence loss, to guide the optimization process of the model. In HU, reconstruction loss measures the discrepancy between original and reconstructed hyperspectral data, expressed mathematically as follows:

$$\mathcal{L}_{\mathrm{rec}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{C}\left(y_{ij} - \hat{y}_{ij}\right)^2 \tag{30}$$

where N denotes the number of pixels, C is the number of spectral bands, y_{ij} represents the true spectral value of pixel i in band j, and $\hat{y}_{ij}$ denotes the corresponding reconstructed value.

The formula for calculating the KL divergence is as follows:

$$\mathcal{L}_{\mathrm{KL}} = \frac{1}{B}\sum_{k=1}^{B}\left[-\frac{1}{2}\sum_{j=1}^{d_z}\left(1 + \log\left(\sigma_{kj}^2\right) - \mu_{kj}^2 - \sigma_{kj}^2\right)\right] \tag{31}$$

where d_z denotes the dimension of the latent space, and μ_{kj} and σ_{kj} represent the mean and variance of the sample kth along the latent dimension jth, respectively.

The total loss function is a weighted sum of the reconstruction loss and the KL divergence loss

$$\mathcal{L}_{\mathrm{VAE}} = \mathcal{L}_{\mathrm{rec}} + \lambda_{\mathrm{KL}} \cdot \mathcal{L}_{\mathrm{KL}} \tag{32}$$

where λ_{KL} is the weighting coefficient for the KL divergence loss, which is used to balance the trade-off between reconstruction accuracy and regularity in latent space.

IV. Experiments

This section evaluates the performance of the proposed method. Three real datasets (Jasper Ridge, Samson, and Apex) were selected for algorithmic experimentation and compared with six different methods in order to validate their effectiveness. The six different methods are UnDIP [32], DeepTrans [23], DGMSSU [33], CyCU-Net [34], PGMSU [35], TCCU-Net [21], and EOT-Net [36]. The optimal results are shown in bold, and the second-best results are underlined. In addition, the impact of each submodule on model performance was evaluated through ablation experiments.

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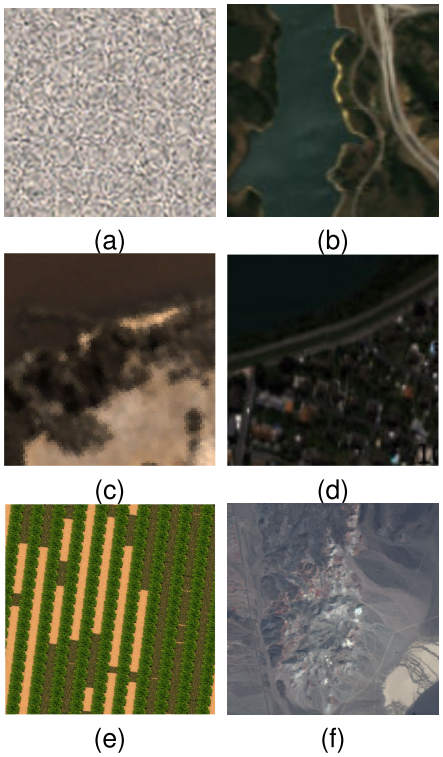


Fig. 4. False-color images on six datasets. (a) False-color image of the Simulated data. (b) False-color image of the Jasper Ridge data. (c) False-color image of the Samson dataset. (d) False-color image of the Apex data. (e) False-color image of the Ray-Tracing dataset. (f) False-color image of the Cuprite data.

A. Data Description

1) Simulate Dataset: This dataset was synthesized based on the spectral characteristics of minerals and vegetation from the U.S. Geological Survey (USGS) spectral library, comprising four endmembers. It has a spatial resolution of 100 × 100 pixels and contains 224 spectral bands covering the range of 0.38–2.50 μm. The abundance value for each pixel is constrained between 0.05 and 0.80 to prevent the presence of pure pixels and to ensure that each pixel is a mixture of at least two endmembers. Following the LMM, the generated abundance matrix was used to mix the endmembers. Finally, white Gaussian noise with a signal-to-noise ratio of 30 dB was added to the mixed pixels. The corresponding color image is shown in Fig. 4(a)

2) Jasper Ridge Dataset: This data was acquired by the AVIRIS sensor over the Jasper Ridge area of California, USA. It has a spatial resolution of 100 × 100 pixels and comprises 224 spectral bands, which cover the spectral range of 0.38–2.50 nm. During subsequent preprocessing, bands heavily affected by water vapor absorption and noise are typically removed, resulting in a final set of 198 effective bands. The scene contains land cover endpoints, such as "roads," "water bodies," "soil," and "trees." The corresponding color image is shown in Fig. 4(b).

3) Samson Dataset: This data was acquired by an airborne sensor over the Port Samson area. The original image has dimensions of 952 × 952 pixels, but a representative subregion of 95 × 95 pixels is typically extracted for analysis in this study. The original spectral data comprises 156 bands spanning wavelengths 400–900 nm and primarily contains three pure endmembers: "soil," "trees," and "water bodies." The corresponding color image is shown in Fig. 4(c).

TABLE I
HYPERPARAMETER SETTINGS FOR DIFFERENT DATASETS

Dataset	lr	KL	batch	z_dim	K	m
Simulate	1e-4	0.001	50	4	400	10
Jasper Ridge	1e-4	0.001	50	4	170	10
Samson	1e-5	0.01	60	3	170	10
Apex	1e-4	0.001	40	4	200	15
Ray-Tracing	1e-4	0.001	50	4	70	10

4) Apex Dataset: This data was acquired by an airborne imaging spectrometer over an urban area in Switzerland. It has a spatial pixel size of 110 × 110 and covers 285 distinct spectral bands, spanning wavelengths 413–2420 nm. The data primarily contains four pure endmembers: "water," "trees," "roads," and "rooftops." The corresponding color image is shown in Fig. 4(d).

5) Ray-Tracing Dataset: This dataset of an orchard scene [37] was generated by utilizing a fully calibrated virtual citrus orchard ray tracer [38]. It has a spatial resolution of 20 × 20 pixels and contains 216 spectral bands covering the wavelength range from 350 to 2500 nm. During subsequent preprocessing, reflectance values from 31 absorption bands were removed, resulting in 185 effective bands retained for analysis. The scene comprises land cover components such as "soil," "weed patches," and "citrus trees." The corresponding color image is presented in Fig. 4(e).

6) Cuprite Dataset: This dataset was acquired by the AVIRIS sensor over the Cuprite mining district in Nevada, USA. It has a spatial resolution of 250 × 190 pixels and contains 224 spectral bands covering the wavelength range from 0.37 to 2.50 μm. During subsequent preprocessing, bands severely affected by noise and atmospheric influences are typically removed, resulting in approximately 188 effective bands retained for analysis. The scene is primarily characterized by typical hydrothermal alteration minerals, including major mineral classes such as "kaolinite," "alunite," "calcite," and "muscovite." The corresponding color composite image is shown in Fig. 4(f).

B. Experimental Setup

1) Parameter Settings: We configured distinct training parameters based on the characteristics of different datasets. Table I provides detailed hyperparameter settings. It is important to note that the number of endmembers for each dataset was set based on prior knowledge. The proposed model does not autonomously estimate this number; rather, the endmember count is treated as a fixed input parameter—a common practice in HU that allows the model to concentrate on optimizing abundance estimation and spectral reconstruction. The experiments were conducted using Python 3.12.17 as the programming language, PyTorch 2.5.1 as the programming

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TABLE II
UNMIXING PERFORMANCE COMPARISON ON THE SIMULATE DATASET

Methods		DeepTrans	DGMSSU	CyCU	PGMSU	UnDIP	TCCU	EOT	Proposed
RMSE	Clay	0.07387	0.20617	0.21459	0.21342	0.24023	0.14185	0.11954	**0.04199**
	Carbonate	**0.06887**	0.33282	0.23121	0.21084	0.36141	0.20066	0.09686	0.10931
	Iron-oxide	0.08287	0.18362	0.37707	0.21204	0.28354	0.15151	0.12548	**0.03092**
	Vegetation	0.10882	0.34711	0.29587	0.21538	0.24171	0.15821	**0.01759**	0.09101
Mean RMSE		0.11127	0.25268	0.27136	0.21538	0.28598	0.15229	0.08986	**0.06831**
SAD	Clay	0.07143	0.39217	0.12201	0.17669	0.12636	0.14016	0.05331	**0.02712**
	Carbonate	0.05629	0.48977	0.06382	**0.01761**	0.10922	0.12539	0.03453	0.03709
	Iron-oxide	0.03601	0.41495	**0.01836**	0.16478	0.11239	0.14445	0.04551	0.02619
	Vegetation	0.02702	0.30813	0.02986	0.14351	0.08347	0.10498	**0.00544**	0.02352
Mean SAD		0.04769	0.40126	0.05851	0.12564	0.10786	0.12874	0.03646	**0.02848**
SRE(db)		19.7	8.74	17.29	12.65	16.31	14.39	20.91	**23.29**

TABLE III
UNMIXING PERFORMANCE COMPARISON ON THE JASPER RIDGE DATASET

Methods		DeepTrans	DGMSSU	CyCU	PGMSU	UnDIP	TCCU	EOT	Proposed
RMSE	Tree	**0.06924**	0.07686	0.09701	0.09763	0.25012	0.08237	0.11335	0.07987
	Water	0.13597	0.08454	0.10521	0.12054	0.23148	0.10753	0.08078	**0.06011**
	Soil	0.19276	0.11221	0.14085	**0.08129**	0.25384	0.11004	0.16234	0.10201
	Road	0.14704	0.07701	0.12557	**0.06588**	0.26841	0.09428	0.10976	0.07271
Mean RMSE		0.14323	0.06634	0.08221	0.07006	0.23225	0.09918	0.12018	**0.06039**
SAD	Tree	0.11513	0.17991	0.04332	0.04861	0.11265	0.07449	0.10109	**0.02815**
	Water	0.22577	0.33443	0.12776	0.19011	0.31352	0.09754	0.13745	**0.04997**
	Soil	0.07573	0.15782	0.05124	**0.02944**	0.12571	0.07225	0.04883	0.06196
	Road	0.06559	0.15931	0.09831	0.03171	0.27866	0.04061	**0.03149**	0.11781
Mean SAD		0.12055	0.20787	0.08016	0.07496	0.20763	0.07122	0.07971	**0.06447**
SRE(db)		16.2	15.1	17.43	18.4	16.82	13.3	16.1	**19.7**

environment, and CUDA version 11.8 for execution. All experiments were conducted on a computer with an NVIDIA GeForce RTX 2070 graphics card with 8 GB of GPU memory.

2) Performance Metrics: This article employs the average root mean square error (aRMSE) metric to evaluate and compare the performance of the model. The aRMSE metric is defined as follows:

$$\text{aRMSE} = \frac{1}{P}\sum_{i=1}^{P}\sqrt{\frac{1}{N}\|\mathbf{a}_i - \hat{\mathbf{a}}_i\|_2^2} \tag{33}$$

where $\mathbf{a}_i$ and $\hat{\mathbf{a}}_i$ represent the true abundance vector and estimated abundance vector of the ith endmembers, respectively.

The average spectral angular distance (aSAD) has been introduced as a measure of the similarity between the ground truth (GT) endmembers and the estimated values. It is defined as follows:

$$\text{aSAD} = \frac{1}{P}\sum_{i=1}^{P}\arccos\left(\frac{\mathbf{m}_i^T\hat{\mathbf{m}}_i}{\|\mathbf{m}_i\|_2\,\|\hat{\mathbf{m}}_i\|_2}\right) \tag{34}$$

where $\mathbf{m}_i$ and $\hat{\mathbf{m}}_i$ represent the true spectral vector and estimated spectral vector of the ith endmembers, respectively.

The signal reconstruction error (SRE) is used to measure the overall difference between the original hyperspectral image and the reconstructed image using the estimated endmembers and abundances. It is calculated as follows:

$$\text{SRE} = 10\log_{10}\left(\frac{\mathbb{E}\left[\|\mathbf{X}\|_F^2\right]}{\mathbb{E}\left[\left\|\mathbf{X}-\hat{\mathbf{X}}\right\|_F^2\right]}\right)\quad(\text{dB}) \tag{35}$$

where $\mathbf{X}$ is the original data matrix, $\hat{\mathbf{X}} = \hat{\mathbf{A}}\hat{\mathbf{S}}$ is the reconstructed data matrix ($\hat{\mathbf{A}}$ is the estimated abundance matrix, $\hat{\mathbf{S}}$ is the estimated endmember matrix), and $\|\cdot\|_F$ denotes the Frobenius norm. Higher SRE values indicate better reconstruction accuracy.

C. Experiments on the Simulate Dataset

The quantitative analysis results for the Simulated dataset are detailed in Table II. The experimental results on the Simulated dataset demonstrate that the proposed method

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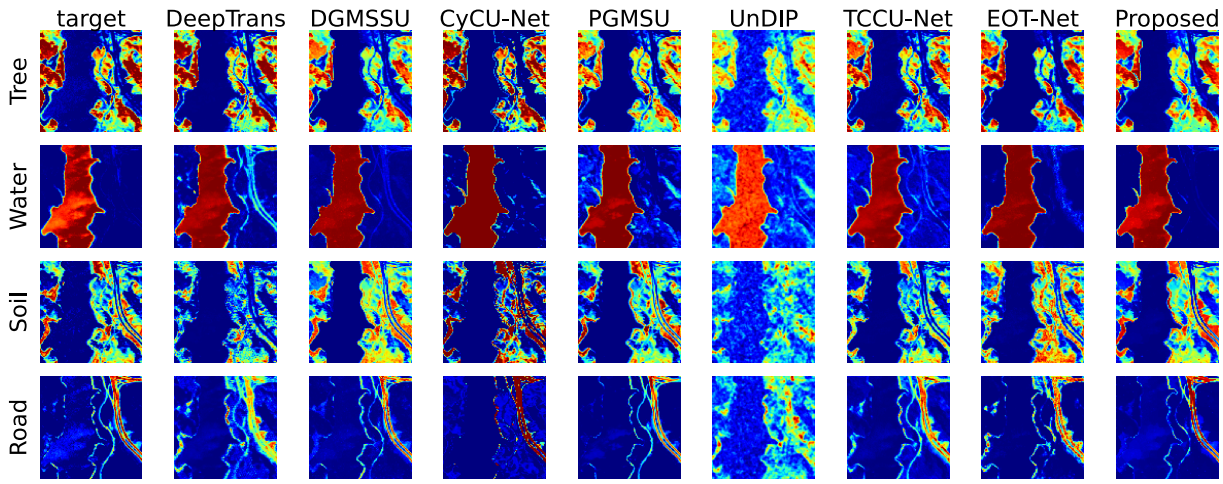


Fig. 5. Abundance map of the Jasper Ridge dataset.

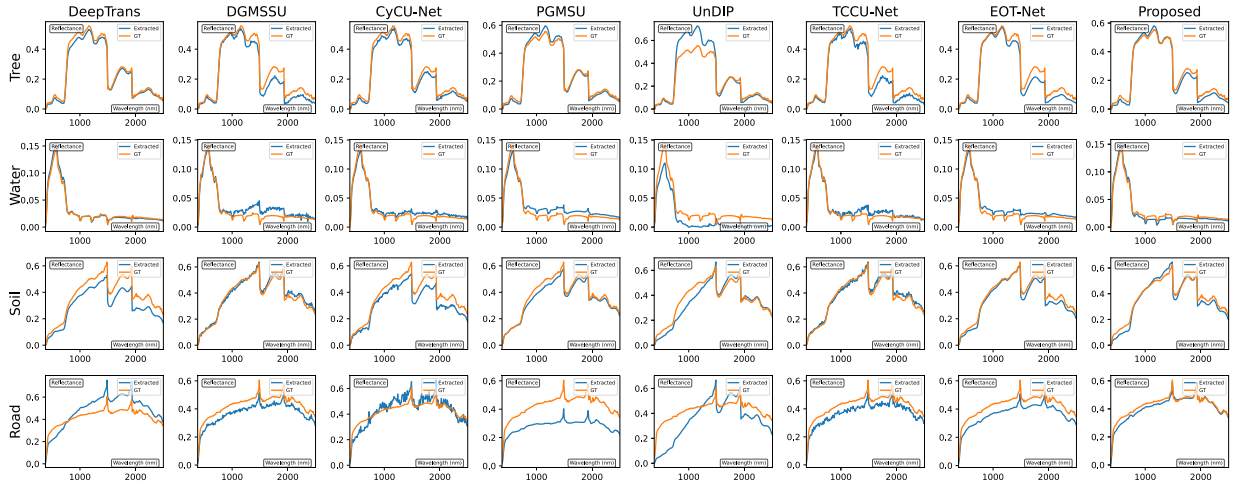


Fig. 6. Endmember graph of the Jasper Ridge dataset.

exhibits superior overall unmixing performance. It achieves better aRMSE, aSAD, and SRE compared to other unmixing methods, validating its comprehensive advantages in both abundance estimation and spectral reconstruction. Although its performance on specific land cover categories (such as carbonate and vegetation) is slightly lower than the single best method for individual metrics, the proposed method maintains stable and high-level performance across all categories with no significant performance weaknesses. This proves its good robustness and generalization capability.

D. Experiments on the Jasper Ridge Dataset

The quantitative results for the Jasper dataset are presented in Table III. As can be seen from the tables, our method outperforms most other approaches in terms of both abundance and endmember estimation. It achieves optimal performance on the mean SAD and mean RMSE metrics, while also delivering competitive results. Figs. 5 and 6 depict the abundance maps and endmember for different methods. The abundance map visualizations show the abundance distribution of the four endmembers (trees, water, soil, and roads) across the rows and the outputs of the various methods across the columns. It can be seen that methods such as DeepTrans and PGMSU exhibit false detections of non-water areas in the "water" endmembers of their abundance maps, indicating limited capability to discriminate between low-reflectance features. The abundance map generated by the UnDIP method exhibits significant spatial blurring and severe loss of edge detail, failing to correctly estimate "roads." Compared to the other six methods, our approach provides the most visually accurate representation of the true abundance map for all four endmembers. It outperforms the other methods in terms of detail recovery, boundary clarity, and feature discrimination accuracy. This further demonstrates the effectiveness of our proposed method compared to other deep learning approaches.

E. Experiments on the Samson Dataset

The quantitative results for the Samson dataset are presented in Table IV. These demonstrate that the proposed model generally outperforms other techniques in terms of abundance

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TABLE IV
UNMIXING PERFORMANCE COMPARISON ON THE SAMSON DATASET

Methods		DeepTrans	DGMSSU	CyCU	PGMSU	UnDIP	TCCU	EOT	Proposed
RMSE	Soil	**0.06799**	0.10639	0.18314	0.17336	0.18451	0.51492	0.12488	0.09194
	Tree	0.07962	0.07551	0.19087	0.15158	0.18452	0.35939	0.10065	**0.07233**
	Water	0.10056	0.10351	0.15556	0.27553	0.18785	0.33313	0.15255	**0.07052**
Mean RMSE		0.08381	0.07755	0.11626	0.14101	0.13543	0.41041	0.12781	**0.06113**
SAD	Soil	0.07322	0.17165	0.03811	0.02162	0.12575	0.01669	0.01589	**0.01283**
	Tree	0.09573	0.13095	0.03643	**0.02681**	0.08021	0.09734	0.07691	0.06789
	Water	0.19017	0.32504	0.14101	0.35931	0.08153	0.07972	0.10943	**0.07407**
Mean SAD		0.11971	0.20921	0.07185	0.13591	0.09631	0.06271	0.06741	**0.05348**
SRE(db)		18.3	17.57	16.91	19.3	17.33	14.81	19.51	**20.32**

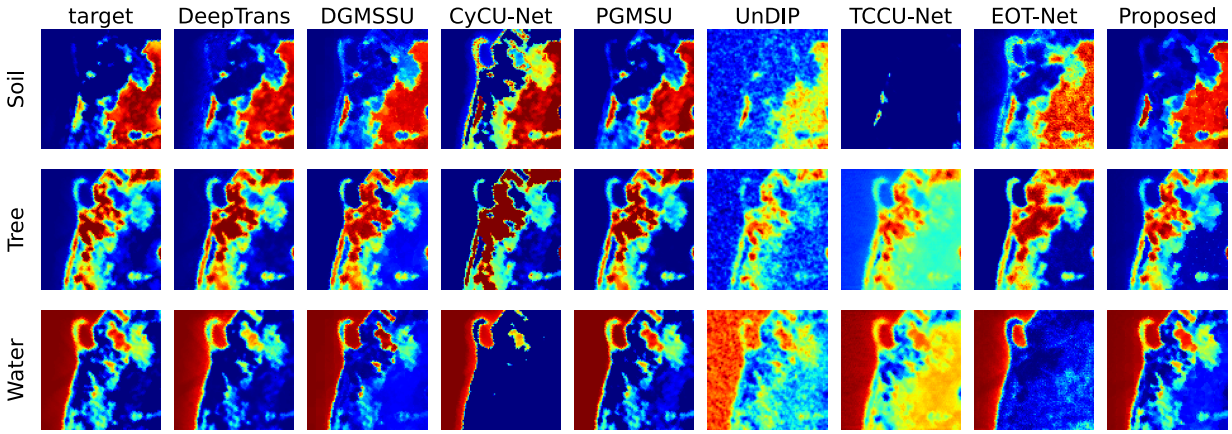


Fig. 7. Abundance map of the Samson dataset.

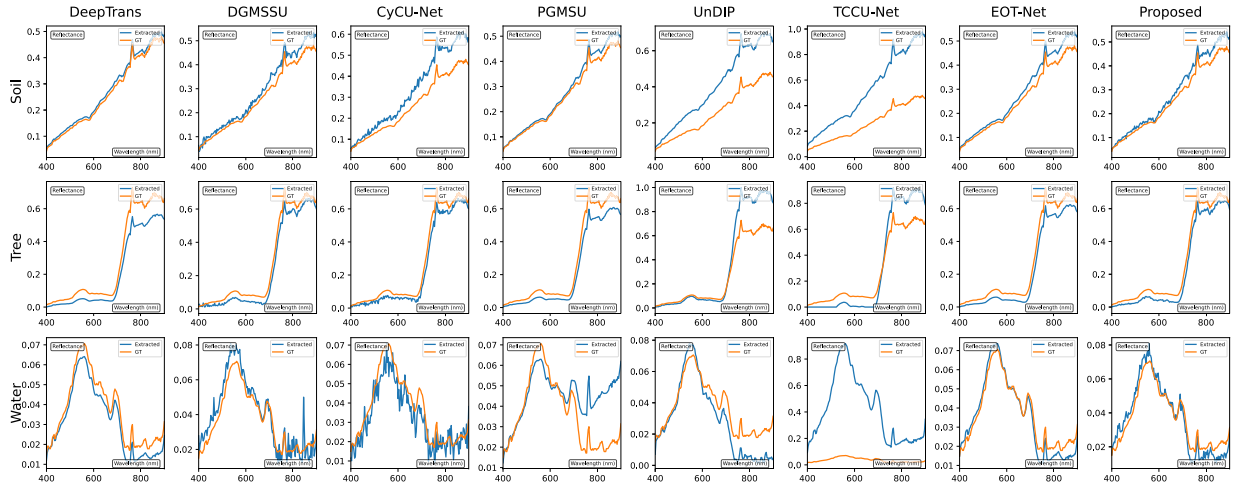


Fig. 8. Endmember graph of the Samson dataset.

and endmember estimation. Figs. 7 and 8 show endmember and abundance maps generated using different methods. It can be seen that approaches such as CyCU-Net utilize recurrent consistency constraints to improve deconvolution performance. However, the results reveal systematic biases in endmember estimation. For the "water" endmember, this method fails to accurately reconstruct the true distribution, resulting in an underestimation of abundance in certain regions. The abundance maps generated by the TCCU-Net method exhibit noticeable spatial blurring, with a severe loss of edge detail

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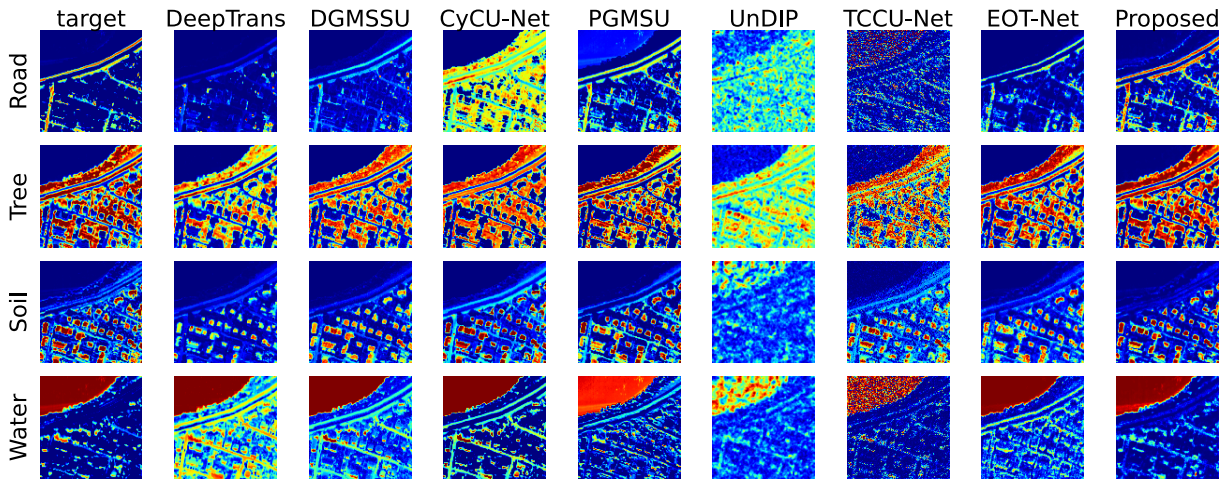


Fig. 9. Abundance map of the Apex dataset.

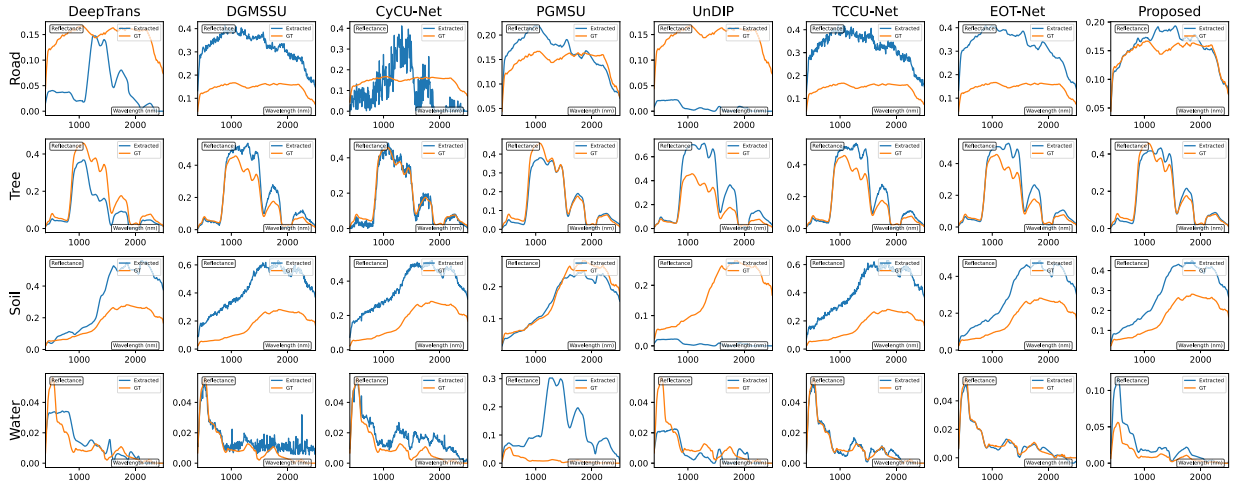


Fig. 10. Endmember graph of the Apex dataset.

TABLE V
UNMIXING PERFORMANCE COMPARISON ON THE APEX DATASET

Methods		DeepTrans	DGMSSU	CyCU	PGMSU	UnDIP	TCCU	EOT	Proposed
RMSE	Road	0.18271	0.16065	0.21344	0.15381	0.33652	0.29497	**0.13547**	0.13786
	Soil	0.15446	0.12551	0.15512	0.15051	0.33411	0.20444	0.15493	**0.09738**
	Tree	0.20779	0.13671	**0.09856**	0.11791	0.33004	0.14201	0.14632	0.11712
	Water	0.33824	0.21652	0.16131	0.18274	0.32769	0.24047	0.22772	**0.08787**
Mean RMSE		0.18904	0.12893	0.12801	0.12491	0.28155	0.22737	0.17002	**0.10901**
SAD	Road	0.15635	0.20874	0.36574	0.14751	0.53792	0.33282	0.13287	**0.08786**
	Soil	0.14591	0.19451	0.11147	0.13172	0.12807	0.22281	0.13446	**0.11022**
	Tree	0.17692	0.22237	0.19112	0.08837	0.21272	0.11036	**0.08062**	0.09846
	Water	0.21116	0.46743	0.36775	0.07131	0.39391	0.33701	0.10574	**0.03983**
Mean SAD		0.17258	0.27326	0.25901	0.12495	0.31668	0.26575	0.11342	**0.07947**
SRE(db)		11.31	14.02	14.94	13.81	16.51	13.35	16.78	**17.32**

and an inability to correctly estimate "soil." This may be due to the limited adaptability of the network structure to complex scenes. Compared to the aforementioned methods, the proposed approach has significant advantages in several

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TABLE VI
UNMIXING PERFORMANCE COMPARISON ON THE RAY-TRACING DATASET

Methods		DeepTrans	DGMSSU	CyCU	PGMSU	UnDIP	TCCU	EOT	Proposed
RMSE	Soil	0.33781	0.19611	0.31347	0.19599	0.32827	0.20278	0.22908	**0.10751**
	Weed	0.31787	0.27397	0.24044	0.17248	**0.11031**	0.29132	0.25571	0.14228
	Tree	0.41481	0.15439	0.25037	0.14525	0.12445	0.19107	**0.09695**	0.21089
Mean RMSE		0.35926	0.20926	0.25668	0.16276	0.25173	0.21551	0.20596	**0.15945**
SAD	Soil	0.25271	0.23644	0.15674	0.20299	0.18772	0.17066	0.15148	**0.07619**
	Weed	0.20377	0.26498	0.11925	0.19602	0.12402	0.20241	0.19407	**0.10069**
	Tree	0.15423	0.14245	0.12533	0.05135	0.08294	0.07915	0.35546	**0.04607**
Mean SAD		0.20357	0.21462	0.13377	0.15012	0.13156	0.15074	0.12700	**0.07432**
SRE(db)		15.54	15.60	11.13	12.22	16.86	12.53	16.93	**18.48**

TABLE VII
COMPARISON OF THE COMPUTATIONAL EFFICIENCY OF DIFFERENT METHODS (TIME: S)

	DeepTrans	**DGMSSU**	**CyCU-Net**	**PGMSU**	**UnDIP**	**TCCU-Net**	**EOT-Net**	**Proposed**
Simulate	13.89	20.93	16.08	171	23.34	21.14	32.65	78.07
Jasper Ridge	14.29	19.04	12.44	162.89	25.03	26.01	32.12	106.62
Samson	13.09	14.49	11.45	141.72	14.16	19.11	25.93	126.36
Apex	17	20.42	18.73	200.62	32.34	31.21	35.38	214.47
ray-tracing	1.96	13.64	3.36	7.64	3	7.41	16.54	97.58

TABLE VIII
ABLATION STUDY OF GGCT-NET MODULE ON THE JASPER RIDGE DATASET. BEST RESULTS ARE HIGHLIGHTED IN BOLD

GCAF	**SCAB**	**aRMSE**	**aSAD**
✗	✗	0.06916	0.12828
✗	✓	0.07322	0.12955
✓	✗	0.06268	0.13836
✓	✓	**0.06039**	**0.06441**

areas. On the Samson dataset, it produces estimates that are closer to true abundances. Visual comparisons confirm its superiority in terms of detail preservation, boundary clarity, and noise suppression. These results provide robust support for the reliability of HU in practical applications.

F. Experiments on the Apex Dataset

The quantitative results of the Apex dataset are presented in Table V. These results suggest that, in most cases, the proposed model outperforms other techniques in terms of abundance and endmember estimation. Figs. 9 and 10 depict the endmember and abundance maps of the various methods. As with the CyCU-Net method, we observed that this method failed to accurately reproduce the true distribution of the "road" endmembers, resulting in false abundance detections in some regions. This may be due to the inadequate optimization of cyclic consistency loss in complex scenes, which leads to poor endmember separation performance. The abundance maps generated by the TCCU-Net and UndIP methods exhibit significant spatial blurring and a severe loss of edge details. They are also unable to accurately estimate "roads" and "trees," which may be related to the limited adaptability of the network structures to complex scenes. In comparison with the aforementioned methods, the abundance map produced by our method in this article demonstrates clear advantages in multiple areas, underscoring the effectiveness of our proposed approach.

G. Experiments on the Ray-Tracing Dataset

The quantitative analysis results for the Ray-Tracing dataset are detailed in Table VI. As shown in the table, the proposed method maintains a leading position in overall unmixing performance, achieving the best scores in aRMSE, aSAD, and SRE. This validates its comprehensive effectiveness in real, complex scenarios. Figs. 11 and 12 present the abundance maps and endmember identification results of different methods, respectively. Although the abundance estimation for the weed and tree categories is slightly inferior to that of the UnDIP and EOT methods, the proposed method achieves the best results in endmember extraction. Furthermore, the endmember spectral comparison plots show that its extracted curves closely match the ground-truth spectra, and its abundance maps reveal clearer spatial distributions of ground objects. This indicates that the proposed method possesses stronger spectral fidelity and spatial decomposition consistency in real data, demonstrating outstanding overall robustness.

H. Experiments on the Cuprite Dataset

The Cuprite dataset was utilized to evaluate the applicability and effectiveness of our method in a real-world scenario. Unlike synthetic datasets, the Cuprite dataset provides only reference spectra for the endmember minerals

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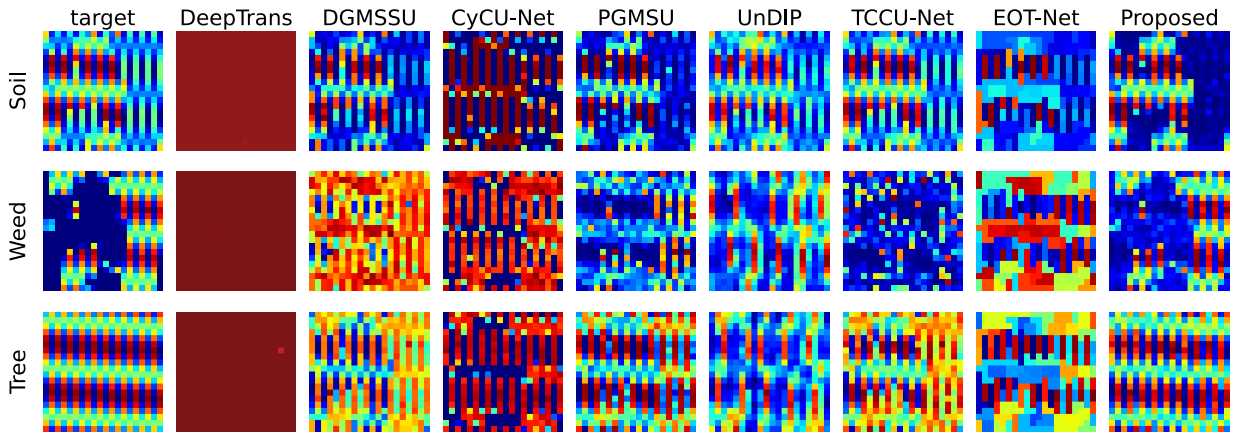


Fig. 11. Abundance map of the Ray-Tracing dataset.

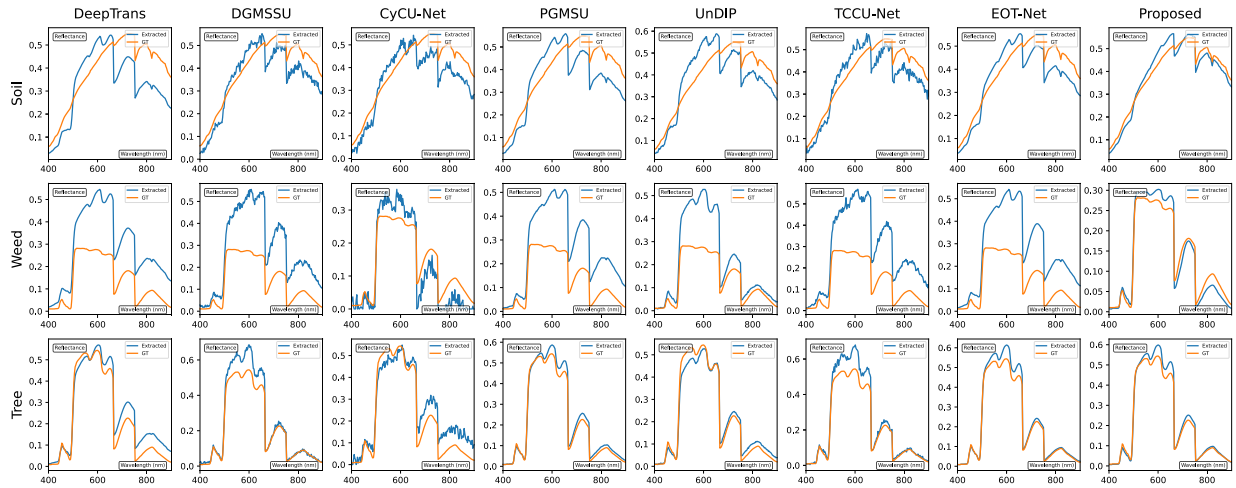


Fig. 12. Endmember graph of the Ray-Tracing dataset.

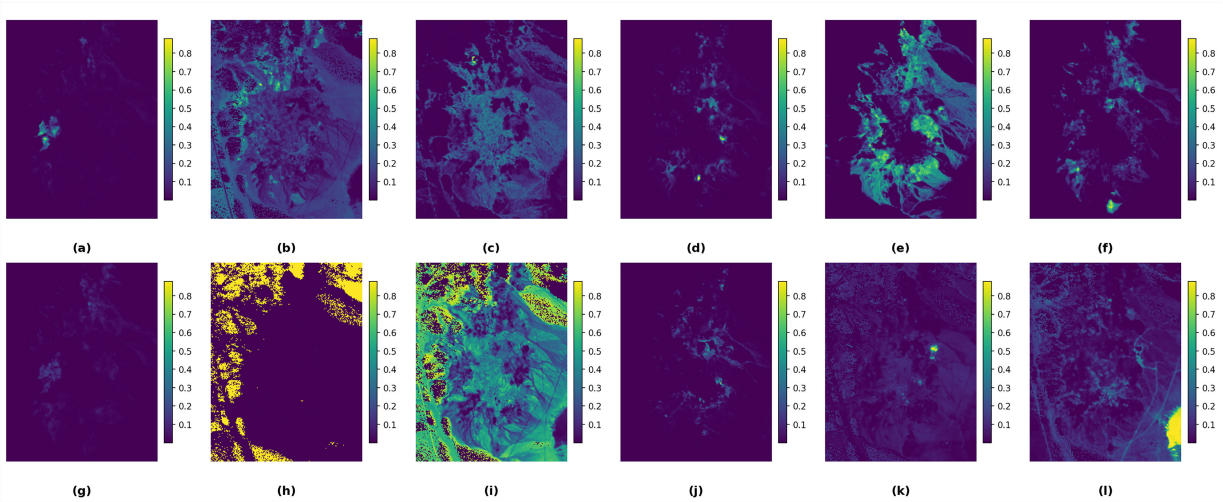


Fig. 13. Abundance maps of the Cuprite dataset for 12 minerals. (a) Buddingtonite GDS85 D-206. (b) Rutile HS189.3B. (c) Andradite WS487. (d) Alunite GDS82 Na82. (e) Kaolin/Smect H89-FR-5 30K. (f) Kaolinite KGa-2. (g) Nontronite NG-1.a. (h) Kaolin/Smect KLF508 85%K. (i) Pyrope WS474. (j) Chalcedony CU91-6A. (k) Muscovite IL107. (l) Montmorillonite + Illi CM37.

and does not contain ground-truth abundance maps. The 12 minerals were primarily identified within this subscene: Alunite GDS82 Na82, Andesite WS487, Buddingtonite GDS85 D-206, Chalcedony CU91-6A, Kaolinite/Smectite

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H89-FR-5 30K, Kaolinite/Smectite KLF508, Kaolinite KGa-2, Montmorillonite+Illite CM37, Muscovite IL107, Albite NG-1.a, Nontronite WS474, and Rutile HS189.3B. The reference spectral signatures were obtained from the U.S. Geological Survey (USGS) spectral library. Fig. 13 presents the abundance maps generated for the Cuprite dataset. The abundance maps clearly illustrate the spatial distribution of the selected minerals, demonstrating the effectiveness of the unmixing process in capturing the material composition of the scene. This strongly validates the applicability and effectiveness of the proposed method in practical applications.

I. Analysis of Computational Efficiency

As shown in Table VII, the proposed GGCT-Net model exhibits higher computational time compared to other unmixing methods. This is primarily attributed to its dual-branch architecture: the G-GCN branch extracts local spatial–spectral correlations guided by superpixels, while the GCA-Transformer branch is designed to model long-range dependencies between spectral and spatial dimensions. This sophisticated mechanism for feature extraction and fusion inevitably introduces additional computational overhead. However, this increased computational cost is compensated by a significant improvement in unmixing performance. These results suggest that, to address the challenges inherent in HU, allocating greater computational resources to construct more powerful models is both necessary and effective.

J. Ablation Study

To evaluate the contributions of the GCAF and SCAB to HU, four model configurations were set up for comparison in the ablation experiments. The metrics are shown in Table VIII. The baseline model, which incorporates no additional modules, reflects fundamental performance. When only the SCAB module was added, the RMSE and SAD values increased slightly, indicating its limited effectiveness when used alone. After incorporating only GCAF, the RMSE value decreased significantly to 0.06268, demonstrating the module's efficacy in feature fusion and information filtering. When both GCAF and SCAB were employed simultaneously, the model achieved optimal performance. RMSE decreased further to 0.06039, while SAD dropped substantially to 0.06447. This demonstrates a synergistic enhancement effect. GCAF effectively integrates features and suppresses redundancy, while SCAB enhances the modeling capabilities of spatial–spectral correlation. Together, they significantly improve demixing accuracy and spectral reconstruction quality.

V. Conclusion

This article presents a dual-branch unmixing model (GGCT-Net), which combines gated graph convolution and GCA-Transformer. The G-GCN branch extracts local spatial–spectral correlations guided by superpixels. The gated cross-attention fusion module enables deep interaction between the endmember and abundance branches, and the spatial–spectral cross-attention module in the GCA-Transformer branch explicitly decouples and models long-range dependencies between the spectral and spatial dimensions. This significantly enhances the accuracy and robustness of the unmixing task. Experimental results on hyperspectral datasets demonstrate that our method outperforms other algorithms across multiple quantitative evaluation metrics. Future research will focus on designing lightweight models to explore their applicability in large-scale hyperspectral scenarios. Furthermore, to enhance the robustness of the framework, we will investigate adaptive or learning-based superpixel generation methods, thereby reducing reliance on predefined clustering algorithms and their associated parameters.

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